

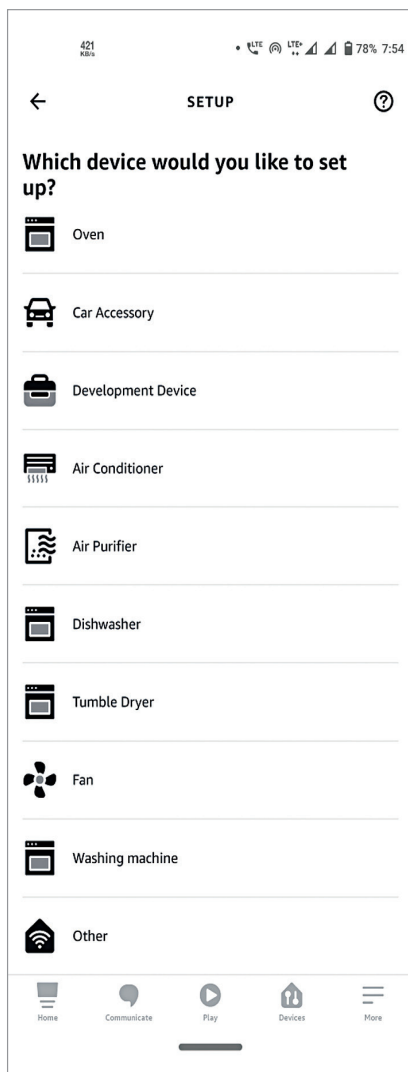


Fig. 15: Alexa app options to add device

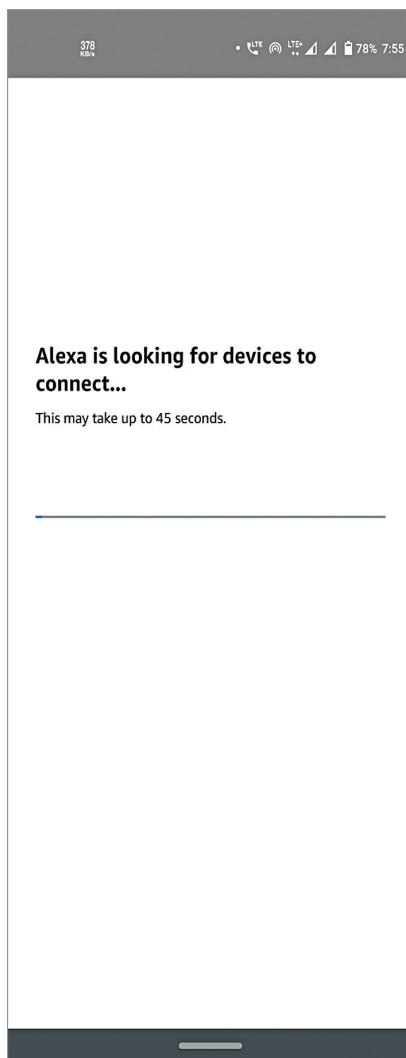


Fig. 16: Alexa searching Wi-Fi device to add on same network

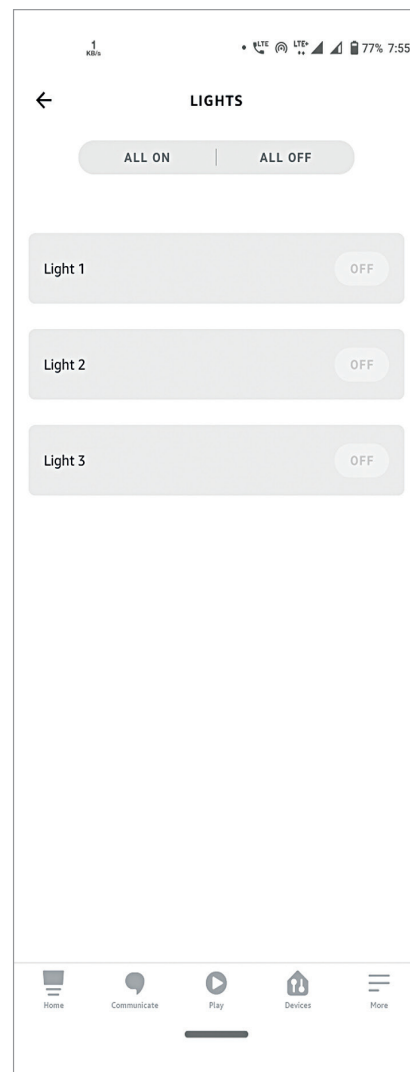


Fig. 17: List device added

Next, create a loop function for checking the commands coming from the serial port. After that, create a nested if condition to check the received commands and define the functions. For instance, if the command is 1 and you have to watch the Discovery channel, set the function for this command to open that particular channel. Then send the channel number of Discovery from IR LED using IR send. That is, if the Discovery channel in your set-top box is 102, then send the code for button 1, then 0, and then 2 with a delay of 200 microseconds between them.

Adding device to Alexa

To add an IoT device to Alexa, open the Alexa app and go to Option→Add Device→Select the device to be

added and then let Alexa search the new device in the network. After the search is completed, a list of devices to be connected is shown such as Light 1, Light 2, Light 3, etc. Select them one by one and add them to Alexa. Now it is ready to control your IoT device(s).

Setting routine

To set the custom voice commands for channels and remote functions to operate routinely, open the Alexa app and go to Settings→Routines→Add new routine. Now set the voice command for routine like “Alexa, Star Plus lagao” (Alexa, select Star Plus) or “Alexa, volume up karo” (Alexa, increase the volume), etc. Then set the action to be performed for that command. Here, Light 1, Light 2, etc will

get toggled according to the button function and command.

Testing

Power the device and ask Alexa (as per the preset command) “Alexa, volume up kar do” (Alexa, raise the volume up) or “Alexa, next channel laga do” (Alexa, switch to the next channel). On hearing the command, Alexa will transfer the action to esp8266, which will further send the command to Arduino and trigger the LED to control the TV as requested.

Congrats! You can now control your TV with Alexa and enjoy your smart home experience. **EFY**

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